Two countable copies of the circle from one

The Open Logic Project contributors

openlogic:set-theory/choice:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author lemma labelled pseudobanachtarski (1036-1040); the proof span 1042-1063 is the author proof environment that immediately follows it.

Editorial note. The unit's expository paragraphs after the proof (recap of the proof strategy, transition to Banach-Tarski) are not part of the counted proof and are excluded.

Editorial note. External reference numbers here are the bare labels (they contain no colons, so no hyphenation is needed).

Editorial scope. One original result of the Open Logic Project unit set-theory/choice, counted once: the pseudo-Banach-Tarski decomposition of the circle. Only the counted lemma and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Rational rotation group and its halves. R is the group of rotations about the origin by rational radian values in [0, 2pi); the unit's disjoint-group lemma partitions R into R_1 (rational radians in [0, pi)) and R_2 = R \ R_1 such that the squares of the elements of either half exhaust R, the map theta to 2 theta being a bijection from [0, pi) to [0, 2pi).. It supplies the two index sets and the reassembly identity {(rho composed with rho) : rho in R_i} = R (1042-1046, 1056-1062).
- Choice set for the circle. C is a set containing exactly one member from each equivalence class of the circle under the relation of differing by a rotation in R, obtained by Choice.. The pieces of the partition are the images rho(C) (1046-1049).
- Cover and no-overlap lemmas. The unit's lemmas: the circle is the union of rho(C) over all rho in R, and distinct rotations give disjoint images of C.. They establish that D_1 and D_2 partition the circle into pairwise disjoint pieces (1049-1053).

Source reference. Original lemma disjointgroup: sources/olp-set-theory-choice/pinned-original-source.tex, source lines 961–961. The cited result is separately proved within the same frozen unit and delivered as its own corpus card; the reference is contextual packaging, and the counted proof retains its own construction and reassembly steps in full.

Source reference. Original lemma vitalicover: sources/olp-set-theory-choice/pinned-original-source.tex, source lines 1012–1012. The cited result is separately proved within the same frozen unit and delivered as its own corpus card; the reference is contextual packaging, and the counted proof retains its own construction and reassembly steps in full.

Source reference. Original lemma vitalinooverlap: sources/olp-set-theory-choice/pinned-original-source.tex, source lines 1022–1022. The cited result is separately proved within the same frozen unit and delivered as its own corpus card; the reference is contextual packaging, and the counted proof retains its own construction and reassembly steps in full.

Lemma 0.1. *There is a partition of S^1 into two disjoint sets, D_1 and D_2, such that D_1 can be partitioned into countably many sets which can be rotated to form a copy of S^1 (and similarly for D_2).*

Proof. Using $\mathrm{SO}(2)_1$ and $\mathrm{SO}(2)_2$ from disjointgroup, let:

$$D_1 = \bigcup_{\rho \in \mathrm{SO}(2)_1} \mathrm{im}(\rho)C \qquad\qquad D_2 = \bigcup_{\rho \in \mathrm{SO}(2)_2} \mathrm{im}(\rho)C$$

This is a partition of S^1, by vitalicover, and D_1 and D_2 are disjoint by vitalinooverlap. By construction, D_1 can be partitioned into countably many sets, $\mathrm{im}(\rho)C$ for each $\rho \in R_1$. And these can be rotated to form a copy of S^1, since $S^1 = \bigcup_{\rho \in \mathrm{SO}(2)} \mathrm{im}(\rho)C = \bigcup_{\rho \in \mathrm{SO}(2)_1} \mathrm{im}((\circ\rho\rho))C$ by disjointgroup and vitalicover. The same reasoning applies to D_2. □

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